

Peng Tang

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EDUCATION

The University of Tokyo

Tokyo, Japan

PhD. In Information and Communication Engineering

Apr. 2026 - Now

- Vector databases, Approximate nearest neighbor search, Learned Index

Zhejiang University

Hangzhou, China

MPhil. In Remote Sensing and Geographic Information System (GPA: 3.88/4.00)

Sep. 2022 - Jun. 2025

- GIS High-Performance Computing Group, Key Lab of Resources & Environmental Information System of Zhejiang Province
- Zhejiang University Award of Honor for Graduate & Key Laboratory Scholarship for Excellence (top 20%)

Zhejiang University

Hangzhou, China

B.Sc. in Geographic Information Science (Major GPA: 4.00/4.00, Overall GPA: 3.69/4.00)

Sep. 2018 - Jun. 2022

- Zhejiang University Outstanding Graduate Title
- Zhejiang University Third-class Scholarship & Professional Scholarship Award for Excellence (top 20%)

Hangzhou No.2 High School of Zhejiang Province

Hangzhou, China

High School

Sep. 2015 - Jun. 2018

RESEARCH EXPERIENCE

Zhejiang University (Key Lab of Resources & Environmental Information System of Zhejiang Province)

Hangzhou, China

Zhejiang Province Student Research Training Project, Advisor: Prof. Zhenhong Du

Mar. 2020 - Jun. 2021

Global-Scale Image Fusion and Visualization Service (Evaluation: Excellent)

- Remote Sensing Image Fusion Processing and Computation Scheme Based on Geotrellis
- Provided Diverse and Customizable Real-Time Rendering Visualization Services

Zhejiang University (Geographic Information Science Major)

Hangzhou, China

Undergraduate Thesis, Advisor: Prof. Feng Zhang

Oct. 2021 - Jun. 2022

Distributed Storage and Computing Methods for Vector Big Data Based on Postgres-XL

- Designed a Distributed Storage Solution for Vector Data Based on Hilbert R-tree & Postgres-XL.
- Designed a High-Performance Computing Framework for Vector Data Based on Spark & GeoTools.
- Four-node parallel retrieval boosts performance by 70%, with a 10% gain from optimized partitioning.

Zhejiang University (Key Lab of Resources & Environmental Information System of Zhejiang Province)

Hangzhou, China

MPhil, Advisor: Prof. Feng Zhang

Spe. 2022 - Jun. 2025

STUBRIN: A Spatio-Temporal Prediction Enhanced Learned Index for Spatial Data (As 1st-Author)

- Published: Tang P, Wang L, Hu L, et al. IEEE Transactions on Knowledge and Data Engineering, 2026.
- Proposed a Spatio-Temporal Update Method to enhance spatial learned indexes using spatio-temporal auto-correlation and future predictions, applied to SLBRIN, and optimized retraining with error threshold and delta learning.
- STUBRIN achieves 1.9-2.4 $\times$, 1.8-13.3 $\times$, 3.4-6.7 $\times$ better build, query and update performance compared to baselines.
- Designed parallel scheduling, which improves build, query and update performance by 6.2-6.8 $\times$, 0.3-4.2 $\times$, 2.6-5.5 $\times$.

An Update and Query Load Optimized Spatial Learned Index with Uniform Partition (As 1st-Author)

- Published: Tang P, Fu C, Hu L, et al. International Journal of Geographical Information Science, 2026.
- Proposed SUPLI, using iterative uniform partitioning and linear regression for efficient index construction.
- Optimized query load and learns from historical queries to enhance efficiency, using a buffer structure for efficient updates.

TBSI: A Transformer-Based Spatial Index for Efficient Construction and Query (As Co-Author)

- Published: Hu Y, Tang P, Meng Y, et al. International Journal of Geographical Information Science, 2025.
- Proposed a novel end-to-end index using an improved quadtree for spatial data organization and uniform partitioning. At each node, a transformer-based model captures and learns data distribution.

DSPI: Spatial Learned Index Leveraging Data-Aware Pivots (As Co-Author)

- Published: Meng Y, Tang P, Hu Y, et al. Transactions in GIS, 2026, 30(2): e70241.

SLBRIN: A Spatial Learned Index Based on BRIN (As Co-Author)

- Published: Wang, Lijun, et al. ISPRS International Journal of Geo-Information 12.4 (2023): 171.
- We proposed a spatial equidistant partition, improving retrieval performance by 0.2-33.8 times vs state-of-the-art methods.

Spatial Learned Index Method for Line Data (Master's Thesis, Advisor: Prof. Feng Zhang)

- Proposed the LIne Learned Index (LILI) for line data and an optimized version, the Graph Grids LIne Learned Index (GG-LILI), using a Graph Grids (G-Grids) intermediate layer.
- Through a graph-based modeling approach, hierarchical indexing algorithms, and intermediate-layer optimizations enhancing spatial feature representation, our methods overcome difficulties in complex spatial object representation and efficient organization.
- LILI improves range query and kNN query performance by 49.1% and 79.2% compared to R*-tree, while GG-LILI reduces storage by 64.9%, enhances static and dynamic query efficiency by 56.9% and 91.3%, and boosts deletion performance by 81.8%.